君正®

Zeratul T23—媒体内存优化说明

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北京君正集成电路股份有限公司
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媒体内存优化说明
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Release history

Date	Revision	Change
Nov.2023	1.0	First release

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1.方案选择

由于不同产品对性能要求不同，比如有的追求极致启动时间，有的追求极小内存占用，所以 T23 zeratul 提供如下两种方案。

方案 A:可达到 t23 极致启动时间，内核阶段可出流。

方案 B:内核阶段不缓存 yuv 数据，400w 分辨率相比方案 A 的 nrvbs 为 2 的情况下，可节省内存。

2.使用说明

方案 A

source 环境时选择 default 配置 un-direct mode

```
please enter the Linux version index(defalt 1): 1
Supported IVDC mode:
  [1] un-direct mode
  [2] direct mode
```

方案 B

source 环境时选择 direct mode

```
please enter the Linux version index(defalt 1): 1
Supported IVDC mode:
  [1] un-direct mode
  [2] direct mode
```

驱动层设置：

1）如截图配置，arch/mips/xburst/lib/isp-t23/include/fast_start_common.h

```
#define Zeratul_Platform  0
```

```
#include <linux/kernel.h>

#define Zeratul_Platform 0              /* 0:dircet 1:undirect*/
#define TX_ISP_FAST_START 1   /* ZRT PLATFORM */
#define TX_ISP_NRVBS 2
#define TX_ISP_FRAME_BUFFER_IGNORE_NUM 1    /* Drop frames in order */
#define FRAME_CHANNEL_DEF_WIDTH 1920
#define FRAME_CHANNEL_DEF_HEIGHT 1080

#define SENSOR_DAY_FPS 15   /* Daytime frame rate */
#define SENSOR_NIGHT_FPS 10 /* Night frame rate */

#define SENSOR_HIGH_FRAME_RATE_MODE 1 /* Sensor High frame rate small image */
#define SENSOR_HIGH_FRAME_RATE_MODE_CHANGE_NUM 5  /* Number of small image frames */

#define DEBUG_TTFF(a) printk("TTFF %s:%d %s:%d\n", __func__, __LINE__, (a==NULL? " " : a), 1000 * *(v

#define K_MODE 1
#define U_MODE 0

#define ALIGN_BLOCK_SIZE (1 << 12)
#define ALIGN_ADDR(base) (((base)+((ALIGN_BLOCK_SIZE)-1))&~((ALIGN_BLOCK_SIZE)-1))
#define ALIGN_SIZE(size) (((size)+((ALIGN_BLOCK_SIZE)-1))&~((ALIGN_BLOCK_SIZE)-1))

char * get_sensor_name(void);
int get_sensor_i2c_addr(void);
int get_sensor_width(void);
int get_sensor_height(void);
INSERT  arch/mips/xburst/lib/isp-t23/include/fast_start_common.h[+]
```

```
#define IR_STATUS_ON "on"
#define IR_STATUS_OFF "off"

#if !Zeratul_Platform
extern int direct_mode;
extern int ivdc_mem_line;
extern int ivdc_threshold_line;

/* 0: un-direct(default configuration)  1: direct */
static int zrt_direct_mode = 1;

/* Configure the number of rows stored in DDR in IVDC (one row represents width, 256 aligned) Y Configure IVDC_ Mem_ Line,
static int zrt_ivdc_mem_line = 2048/2;

/* 0: configured as height/2, indicating a threshold of half a frame (default value) */
static int zrt_ivdc_threshold_line = 0;
#endif

enum IR_STATUS{
    IR_STATUS_DAY = 0,  /* day mode */
    IR_STATUS_NIGHT = 1, /* night mode */
NORMAL  arch/mips/xburst/lib/isp-t23/tx-isp-fast.c                    c      utf-8      4%
```

2）rm arch/mips/xburst2/lib/isp-t23/lib.a 后，重新编译内核烧录。

3.直通/非直通模式下次码流缓存

（1）非直通模式下想用次码流缓存

```
if [ ${ZRT_ENV_SOC_TYPE} = SOC_T23Z ]; then
    if [ ${ZRT_ENV_FLASH_TYPE} = NOR ]; then
        if [ ${ZRT_ENV_SENSOR_NUM} = one ]; then
            CMDLINE="console=${UART},115200n8 mem=32M@0x0 rmem=32M@0x2000000 root=/dev/ram0 rw rdinit=/linux
            ##### mem 系统内存分配20m启动配置 ######
            #    CMDLINE="console=${UART},115200n8 mem=20M@0x0 rmem=46M@0x1400000 root=/dev/ram0 rw rdinit=/
            BOOTINFO="kernel=5M@0x98000 rootfs=3M@0x598000"
            ENVINFO="senv;[HW];init_vw=1280;init_vh=720;nrvbs=2;init_vw_ch1=640;init_vh_ch1=360;nrvbs_ch1=2;
            #ENVINFO="senv;[HW];init_vw=1280;init_vh=720;nrvbs=2;mode=0;pwm_duty=20;pwm_freq=80000;eenv;"
        else
            CMDLINE="console=${UART},115200n8 mem=32M@0x0 rmem=32M@0x2000000 root=/dev/ram0 rw rdinit=/linux
            BOOTINFO="kernel=5M@0xC8000 rootfs=3M@0x5C8000"
            ENVINFO="senv;[HW];init_vw=1920;init_vh=1080;nrvbs=1;sub_init_vw=1920;sub_init_vh=1080;sub_nrvbs=
        fi
    else
        if [ ${ZRT_ENV_SENSOR_NUM} = one ]; then
            CMDLINE="console=${UART},115200n8 mem=36M@0x0 rmem=28M@0x2400000 root=/dev/ram0 rw rdinit=/linux
            BOOTINFO="kernel=4M@0x200000 rootfs=4M@0x600000"
            ENVINFO="senv;[HW];init_vw=2048;init_vh=1536;nrvbs=2;mode=0;pwm_duty=20;pwm_freq=80000;eenv;"
        else
            CMDLINE="console=${UART},115200n8 mem=36M@0x0 rmem=28M@0x2400000 root=/dev/ram0 rw rdinit=/linux
            BOOTINFO="kernel=4M@0x200000 rootfs=4M@0x600000"
            ENVINFO="senv;[HW];init_vw=1920;init_vh=1080;nrvbs=1;sub_init_vw=1920;sub_init_vh=1080;sub_nrvbs=
INSERT  tools/make_tag/tag_generator/build_tag_t23.sh[+]
```

在 build_tag_t23.sh 文件中，新增

```
init_vw_ch1=640;init_vh_ch1=360;nrvbs_ch1=2;
```

标签，init_vw_ch1 表示次码流的宽，init_vh_ch1 表示次码流的高，nrvbs_ch1 表示次码流要在内核阶段缓存的帧数。配置好之后重新编译 tag.bin 固件并烧录到设备上。

```
[root@Zeratul:tmp]# dmesg|grep TTFF
[    0.000000] TTFF tx_isp_riscv_hf_prepare:395 tx_isp_stop_riscv start:202
[    0.000000] TTFF tx_isp_riscv_hf_prepare:416 wait riscv frame done:227
[    0.000000] TTFF tx_isp_riscv_hf_prepare:428 tx_isp_stop_riscv done:232
[    0.000000] TTFF frame_channel1_fast_start 148 W:640 H:360 N:2
[    0.012399] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 0 Start:272
[    0.045469] TTFF frame_channel_buffer_done:2177 Chn:1 Buf:0 Write Done:305
[    0.045712] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 1 Start:305
[    0.078904] TTFF frame_channel_buffer_done:2177 Chn:1 Buf:1 Write Done:338
[    0.079049] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 2 Start:338
[    0.112386] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 3 Start:372
[    0.145720] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 4 Start:405
[    0.179059] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 5 Start:438
[    0.212395] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 6 Start:472
[    0.245724] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 7 Start:505
[    0.279063] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 8 Start:538
[    0.312396] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 9 Start:572
```

设备启动之后，输入 dmesg|grep TTFF 命令，可以查看次码流缓存到内核成功。

（2）直通模式下想用次码流缓存

```
        if [ ${ZRT_ENV_SENSOR_NUM} = one ]; then
            CMDLINE="console=${UART},115200n8 mem=32M@0x0 rmem=32M@0x2000000 root=/dev/ram0 rw rdinit=/linuxrc mtdparts=jz_sfc:256K(boot),352K(
            ##### mem 系统内存分配20m启动配置 ######
            #    CMDLINE="console=${UART},115200n8 mem=20M@0x0 rmem=46M@0x1400000 root=/dev/ram0 rw rdinit=/linuxrc mtdparts=jz_sfc:256K(boot),
            BOOTINFO="kernel=5M@0x98000 rootfs=3M@0x598000"
            ENVINFO="senv;[HW];init_vw=1280;init_vh=720;nrvbs=2;init_vw_ch1=640;init_vh_ch1=360;nrvbs_ch1=2;direct_mode_save_kernel_ch1=0;mode=0
            #ENVINFO="senv;[HW];init_vw=1280;init_vh=720;nrvbs=2;mode=0;pwm_duty=20;pwm_freq=80000;eenv;"
        else
            CMDLINE="console=${UART},115200n8 mem=32M@0x0 rmem=32M@0x2000000 root=/dev/ram0 rw rdinit=/linuxrc mtdparts=jz_sfc:256K(boot),544K(
            BOOTINFO="kernel=5M@0xC8000 rootfs=3M@0x5C8000"
            ENVINFO="senv;[HW];init_vw=1920;init_vh=1080;nrvbs=1;sub_init_vw=1920;sub_init_vh=1080;sub_nrvbs=1;mode=0;sensor_num=${SENSOR_NUM};e
        fi
    else
        if [ ${ZRT_ENV_SENSOR_NUM} = one ]; then
            CMDLINE="console=${UART},115200n8 mem=36M@0x0 rmem=28M@0x2400000 root=/dev/ram0 rw rdinit=/linuxrc mtdparts=sfc_nand:1M(boot),1M(tag
            BOOTINFO="kernel=4M@0x200000 rootfs=4M@0x600000"
            ENVINFO="senv;[HW];init_vw=2048;init_vh=1536;nrvbs=2;mode=0;pwm_duty=20;pwm_freq=80000;eenv;"
        else
            CMDLINE="console=${UART},115200n8 mem=36M@0x0 rmem=28M@0x2400000 root=/dev/ram0 rw rdinit=/linuxrc mtdparts=sfc_nand:1M(boot),1M(tag
            BOOTINFO="kernel=4M@0x200000 rootfs=4M@0x600000"
            ENVINFO="senv;[HW];init_vw=1920;init_vh=1080;nrvbs=1;sub_init_vw=1920;sub_init_vh=1080;sub_nrvbs=1;mode=0;sensor_num=${SENSOR_NUM};e
SERT  tools/make_tag/tag_generator/build_tag_t23.sh[+]                                    sh    utf-8    26%    25/96

INSERT --
```

在 build_tag_t23.sh 文件中，新增

```
init_vw_ch1=640;init_vh_ch1=360;nrvbs_ch1=2;direct_mode_save_kernel_ch1=1;
```

标签，init_vw_ch1 表示次码流的宽，init_vh_ch1 表示次码流的高，nrvbs_ch1

表示次码流要在内核阶段缓存的帧数，direct_mode_save_kernel_ch1=1 表示启用次码流缓存，不写这个标签或者这个值等于 0 将不启用次码流缓存。配置好之后重新编译 tag.bin 固件并烧录到设备上。

```
[root@Zeratul:tmp]# dmesg|grep TTFF
[    0.000000] TTFF tx_isp_riscv_hf_prepare:395 tx_isp_stop_riscv start:202
[    0.000000] TTFF tx_isp_riscv_hf_prepare:416 wait riscv frame done:227
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[    0.000000] TTFF frame_channel1 fast start 148 W:640 H:360 N:2
[    0.012399] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 0 Start:272
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[    0.045712] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 1 Start:305
[    0.078904] TTFF frame_channel_buffer_done:2177 Chn:1 Buf:1 Write Done:338
[    0.079049] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 2 Start:338
[    0.112386] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 3 Start:372
[    0.145720] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 4 Start:405
[    0.179059] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 5 Start:438
[    0.212395] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 6 Start:472
[    0.245724] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 7 Start:505
[    0.279063] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 8 Start:538
[    0.312396] TTFF isp_vic_interrupt_service_routine0:956 Chn:0  Frame 9 Start:572
```

设备启动之后，输入 dmesg|grep TTFF 命令，可以查看次码流缓存到内核成功。